Containing Epidemics on the Global Air-Transport Network: Budget, Centrality, and the Epidemic Threshold

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TABLE I
OpenFlights network summary

Measure	Value
Airports (full graph)	3,214
Routes (full graph)	18,858
Largest-component airports	3,188
Largest-component routes	18,833
Average degree (largest component)	11.81
Connected components	7

Abstract—We study how limited airport closures affect epidemic spreading on a network derived from global air-transport routes. The analysis combines centrality-based intervention strategies, Euler-integrated SIR dynamics, and network resilience measurements. Airports are ranked using degree, closeness, betweenness, and eigenvector centrality, and the resulting strategies are compared with random closure under equal budgets.

I. Motivation / Problem Statement

Air travel connects distant populations and can accelerate the geographic spread of epidemics. Since closing all airports is impractical, this work asks which airports should be closed under a limited budget. [1], [2]

II. Methodology

A. Network construction

We use the OpenFlights airport and route files downloaded on 27 October 2026 [3]. Airports are nodes and routes are undirected edges. Invalid numeric identifiers, self-loops, duplicate routes, and isolated airports are removed. The cleaned graph has 3,214 airports and 18,858 connections in seven components; its largest component has 3,188 airports and 18,833 connections. Table I summarizes the graph used for the epidemic experiments.

B. Centrality and interventions

We calculate degree, closeness, betweenness, and eigenvector centrality using NetworkX, NumPy, pandas, and Matplotlib [4], [5], [6], [7]. For a budget b, the top $\lceil b|V| \rceil$ airports are removed for a centrality strategy. Random closure samples the same number of airports using a fixed seed.

TABLE II
Top airports by centrality in the largest component

Measure	Top three airports (OpenFlights ID)
Degree	Schiphol (580), Frankfurt (340), Charles de Gaulle (1382)
Closeness	Frankfurt (340), Charles de Gaulle (1382), Heathrow (507)
Betweenness	Charles de Gaulle (1382), LAX (3484), Anchorage (3774)
Eigenvector	Schiphol (580), Frankfurt (340), Charles de Gaulle (1382)

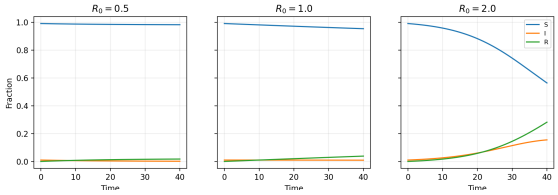


Fig. 1. Mean-field SIR behavior below, at, and above the epidemic threshold.

C. SIR threshold

The normalized mean-field model is

$$\dot{S} = -\beta SI, \quad \dot{I} = \beta SI - \gamma I, \quad \dot{R} = \gamma I. \tag{1}$$

The threshold is $R_0 = \beta S_0/\gamma$. The disease-free equilibrium is stable below $R_0 = 1$ and loses stability at the transcritical threshold.

D. Network metapopulation model

Each airport has susceptible, infected, and recovered populations. Local transmission is $\beta S_i I_i / N_i$, recovery is γI_i, and a graph-diffusion term couples neighboring airports. Euler integration is used with $\beta = 0.30$, $\gamma = 0.10$, mobility rate 0.02, normalized population one per airport, $dt = 0.1$, and duration 80. The initial infected fraction is 1% at one fixed high-degree seed airport. The seed airport is protected from closure so that every strategy is evaluated from the same initial condition.

E. Interventions and measurements

We compare random, degree, closeness, betweenness, and eigenvector closures for budgets from 1% to 20%. Outcomes include final outbreak size, peak infection, time to peak, and largest connected component. Final outbreak

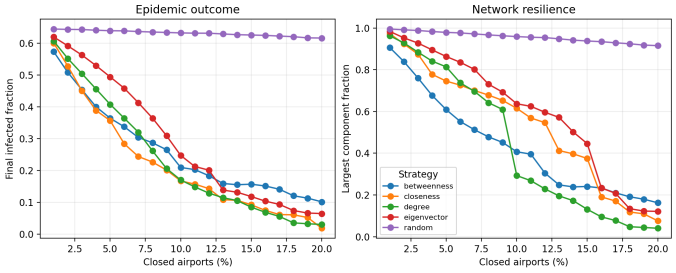


Fig. 2. Complete 1–20% closure-budget results on the largest OpenFlights component.

TABLE III
Final outbreak and connectivity at a 20% budget

Strategy	Final outbreak	Peak infected	LCC fraction
Random	61.63%	25.45%	91.55%
Degree	3.00%	0.80%	4.04%
Closeness	1.80%	0.59%	7.57%
Betweenness	10.11%	3.22%	16.24%
Eigenvector	6.43%	2.22%	12.12%

size is the cumulative recovered fraction at the end of the simulation. We define containment as final outbreak size at most 5%. The complete sweep evaluates every integer budget from 1% to 20%; random closure uses five independent trials.

III. Results and Key Findings

Figure 2 shows the complete generated real-network comparison. At a 20% budget, closeness has the smallest final outbreak size (1.80%), followed by degree (3.00%), eigenvector (6.43%), betweenness (10.11%), and random closure (61.63%). Degree reaches the 5% containment criterion first, at 18while betweenness, eigenvector, and random closure do not reach it within the tested range. Targeted strategies reduce outbreaks more strongly, but they also fragment the network more severely.

The compact synthetic comparison at a 10% budget produced final outbreak fractions of 0.28–0.29% for ER, 0.28% for BA, and 0.28% for WS over the 20-time-unit validation run. The degree-preserving null comparison retained almost the same random-closure connectivity as the real network (96.5% versus 96.610%), while degree-based removal produced 29.3% for the real graph and 61.5% for the randomized graph. This indicates that detailed topology, not only the degree sequence, affects fragmentation.

Delayed intervention was evaluated with a fixed 10% betweenness closure budget. With immediate, 20-unit, and 40-unit intervention, final outbreak sizes were 18.83%, 54.15%, and 57.02peak infected fractions were 6.30%, 23.21%, and 24.44%. Thus, intervening later substantially increases outbreak size in this model.

IV. Discussion / Analysis

The real network is heterogeneous: a small set of airports has a large role in connecting otherwise weakly connected regions. Degree and closeness performed best for reducing the cumulative outbreak under the tested parameters, whereas betweenness caused strong fragmentation but did not achieve the 5% containment criterion by 20%. This demonstrates that structural importance for connectivity and epidemic suppression need not be identical. The appropriate strategy depends on whether the objective prioritizes epidemic reduction, connectivity preservation, or a weighted combination of both.

The model has several limitations. OpenFlights route records do not provide passenger populations or time-varying traffic, so normalized airport populations and unweighted routes are used. The SIR parameters are scenario parameters rather than estimates for a particular disease. Finally, the intervention comparison measures structural removal and does not model rerouting or behavioral adaptation.

V. Conclusion

This project provides a reproducible framework for comparing airport closure strategies on a real flight network. The mean-field model validates the epidemic threshold, while the metapopulation model connects local SIR dynamics to global network structure. The final parameter sweep will identify the smallest budget satisfying the stated containment criterion and quantify its connectivity cost.

References

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